Alan Bao

858-733-9696 | alanb@berkeley.edu | linkedin.com/in/alan-bao | alanbao.org

Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science, Electrical Engineering and Computer Sciences — *May 2026*

- **GPA**: 3.71
- **Relevant Coursework**: Data Structures, Algorithms, Computer Architecture, Databases, Computer Security, Internet Processes, Machine Learning, Robotics, Probability, Optimization, Numerical Analysis

Experience

Software Engineer Intern — June 2025 – Aug. 2025
Capital One — *Plano, TX*

- Expedited auto-loan approval process by 23% by deploying an accessible graph-based auto-loan decisioning service
- Enabled reactive graph editing in the UI by building a secure REST API endpoint that processes rule files, logs graph and versioning metadata and persists them to S3 and Postgres
- Reduced graph build time by 66% and rebuild time by 14% by designing an optimized AWS Postgres (RDS) schema for necessary graph data and implementing lightweight incremental rebuilds
- Decreased data retrieval time from 8 to 3 seconds by integrating a cache to eliminate redundant queries
- Shrank JavaScript bundle size by 37% by migrating the UI from Angular to Lit

Data Science Intern — Feb. 2023 – Dec. 2024
Rapid Reviews Infectious Diseases — *Berkeley, CA*

- Increased review efficiency by 41% by building a classifier that auto-assigns preprints to reviewers by topic
- Delivered weekly trend briefs to 20 reviewers by mining and analyzing 500+ papers for trending research topics
- Streamlined the review processes by manually matching 40-80 reviewers to preprints per week

Software Engineer Intern — May 2024 – Aug. 2024
Hologic — *San Diego, CA*

- Boosted analytics throughput 70% by architecting the News Aggregation Program (NAP), an automated text mining and classification pipeline that processes 2000+ medical articles per week
- Improved UX and halved article search time to 4 seconds by redesigning NAP's UI and optimizing database queries
- Halved research time and identified key priorities using topic modeling on 5 years of medical literature
- Built a low latency, company-wide email digest system, automatically sending 100+ targeted summaries biweekly

Projects

Cooperative Drone Autonomy Platform | *Python, C++, ROS2, OpenCV* — Sept. 2025 – Dec. 2025

- Built a distributed ROS2 system for perception, planning and control in a two-drone autonomous delivery system
- Enabled autonomous navigation with 0.5-meter payload accuracy by implementing a perception pipeline to create navigable maps synchronously communicated between both drones

Python IPv4 Networking Protocols | *Python, Socket Programming* — Jan. 2025 – May 2025

- Built a Python networking stack implementing traceroute (UDP/ICMP), distance-vector routing with loop prevention and a TCP-style reliable transport
- Designed reliability mechanisms including sliding windows, retransmissions and adaptive RTT/RTO estimation

UC Berkeley Assist Tool | *Python, React, JavaScript, Selenium, HTML, CSS* — May 2024 – Aug. 2025

- Collected 6000+ course articulations in under 14 minutes by automating data collection through web scraping
- Implemented a React web application that helps students instantly research transferable courses from all 116 California Community Colleges to UC Berkeley

Technical Skills

Languages: Python, Java, Golang, C/C++, C#, JavaScript, SQL, HTML/CSS, RISC-V, x86
Frameworks: React, Node.js, Spring Boot, REST APIs
Developer Tools: Docker, AWS, PostgreSQL, Git, Linux, CI/CD, LLMs, Agile
Libraries: PyTorch, Numpy, Pandas, NLTK, Pytest, JUnit, Socket Programming